

Program-1:- Development of a Basic 3D Scene in Unity with Player Movement

Objective: Develop a scenario in which two game objects are there.

–One game object is having the capability of the movement corresponding to upward & downward, left & right, forward & backward.

–The other game object should start as a platform.

Develop a game in which the capsule should be placed on the platform (rectangular cube) composing of all the operation mentioned above.



Software & Tools Used

- Unity Hub
 - Unity Editor (3D Core Template)
 - Microsoft Visual Studio
 - Windows Operating System
-



Theory

Unity is a real-time 3D development engine used to create games, AR/VR applications, and simulations.

In Unity:

- A Game Object is any object present in the scene (Cube, Capsule, Camera, etc.).
- Every Game Object has a Transform Component that controls its Position, Rotation, and Scale.
- A Rigidbody Component provides physics properties.
- C# scripting is used to define object behavior.

In this program:

- A Cube is used as a rectangular platform.
- A Capsule is used as the moving object.

- Movement is implemented using keyboard input.
 - The `Update()` method continuously checks for key presses.
 - The `transform.Translate()` function moves the capsule in 3D space.
-

Procedure

Step 1: Create New Project

1. Open Unity Hub.
 2. Click New Project.
 3. Select 3D (Core) template.
 4. Name the project: `Lab1_ARVR`.
 5. Click Create.
-

Step 2: Create Platform (Rectangular Cube)

1. In Hierarchy → Right Click → 3D Object → Cube.
2. Rename it as Platform.
3. In Inspector → Transform:

Position:

X = 0

Y = 0

Z = 0

Scale:

X = 5

Y = 1

Z = 5

This creates a flat rectangular platform.

Step 3: Create Capsule (Moving Object)

1. Right Click → 3D Object → Capsule.
2. Rename it as Player.
3. Set Position:

X = 0

Y = 2

Z = 0

The capsule is placed above the platform.

Step 4: Add Rigidbody Component

1. Select Player.
2. Click Add Component.
3. Add Rigidbody.
4. Uncheck Use Gravity.

This prevents the capsule from falling due to gravity.

Step 5: Create Movement Script

1. In Project Window → Right Click → Create → C# Script.
2. Name it PlayerMovement.
3. Open the script in Visual Studio.
4. Replace the default code with the following:

```
using UnityEngine;
```

```
public class PlayerMovement : MonoBehaviour
```

```
{
```

```
    public float speed = 5f;
```

```
    void Update()
```

```
    {
```

```
        float moveX = 0;
```

```
        float moveY = 0;
```

```
        float moveZ = 0;
```

```
        // Left & Right
```

```
        if (Input.GetKey(KeyCode.A))
```

```
        {
```

```
            moveX = -1;
```

```
        }
```

```
        if (Input.GetKey(KeyCode.D))
```

```
        {
```

```
            moveX = 1;
```

```
}
```

```
// Forward & Backward
```

```
if (Input.GetKey(KeyCode.W))
```

```
{
```

```
    moveZ = 1;
```

```
}
```

```
if (Input.GetKey(KeyCode.S))
```

```
{
```

```
    moveZ = -1;
```

```
}
```

```
// Up & Down
```

```
if (Input.GetKey(KeyCode.E))
```

```
{
```

```
    moveY = 1;
```

```
}
```

```
if (Input.GetKey(KeyCode.Q))
```

```
{
```

```
    moveY = -1;
```

```
}
```

```
Vector3 movement = new Vector3(moveX, moveY, moveZ);  
  
transform.Translate(movement * speed * Time.deltaTime);  
  
}  
  
}
```

5. Save the script.
6. Drag and drop the script onto the Player object.

Now the capsule can move in six directions.

Step 6: Adjust Camera

Select Main Camera and set:

Position:

X = 0

Y = 8

Z = -10

Rotation:

X = 30

Y = 0

Z = 0

This gives a proper view of the platform and capsule.

Step 7: Execution

1. Click the Play button.
2. Use the following keys:

Key	Action
-----	--------

W	Forward
---	---------

S	Backward
---	----------

A	Left
---	------

D	Right
---	-------

E	Up
---	----

Q	Down
---	------

3.
Move the capsule and place it properly on the rectangular platform.

 **Working**

- The `Update()` method runs every frame.
 - `Input.GetKey()` checks whether a specific key is pressed.
 - A movement vector is created using X, Y, and Z values.
 - `transform.Translate()` moves the capsule relative to its current position.
 - By controlling movement in all six directions, the capsule is positioned on the platform.
-



Observation

- The capsule responds correctly to all directional inputs.
 - Smooth real-time movement is achieved.
 - The capsule can be successfully placed on the rectangular cube platform.
-



Result

The required 3D scene was successfully developed.

Two game objects were created, and the capsule was moved in upward, downward, left, right, forward, and backward directions and placed on the rectangular platform as specified.



Conclusion

The program was successfully implemented in Unity.

Six-directional movement of a capsule was achieved using C# scripting, and the objective of placing it on a rectangular cube platform was accomplished.



Viva Questions

Q1: Why did you use Rigidbody?

A: To allow physics properties and smoother movement control.

Q2: What is transform.Translate()?

A: It is a method used to move an object in 3D space relative to its current position.

Q3: What is a Game Object?

A: Any object present in the Unity scene is called a Game Object.

How To Recreate Later Create 3D project

- 1. Add Cube → Resize**
- 2. Add Capsule → Position above**
- 3. Add Rigidbody → Disable Gravity**
- 4. Write Movement Script**
- 5. Attach Script**
- 6. Adjust Camera**
- 7. Improve visuals**